Cascading Style Sheets (CSS) (Part 1)

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Agenda

- What is CSS?
- CSS Syntax
- CSS Basic

What is CSS?

- **CSS** stands for **C**ascading **S**tyle **S**heets
- Styles define **how to display** HTML elements
- Styles are normally stored in **Style Sheets**
- Multiple style definitions will **cascade** into one

Why Style Sheets?

- Style sheets define How HTML elements are to be displayed
- Styles are normally saved in external .css files
- External style sheets enable you to change the appearance and layout of all the pages in your Web
  - By editing only one single CSS document

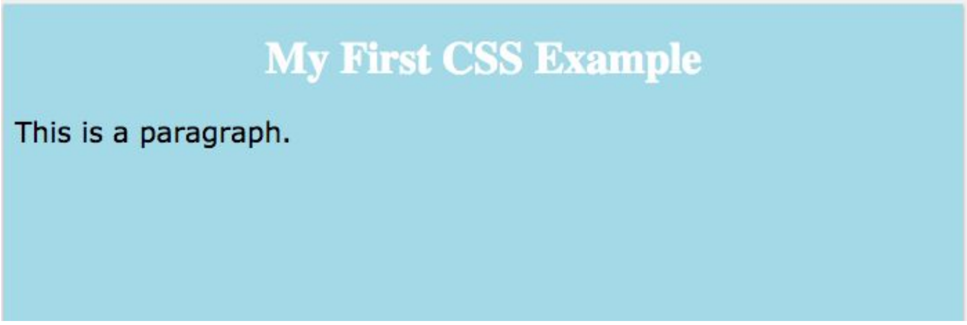
Multiple styles will cascade into one

- Style sheets allow style information to be specified in many ways.
- Styles can be specified:
  - Inside an HTML element
  - Inside the head section of an HTML page
  - In an external CSS file
- Cascading order - What style will be used when there is more than one style specified for an HTML element?

Example: e1.html

e1.html (css) — Brackets

```html
<!DOCTYPE html>
<html>
<head>
</head>
<body>

<h1>My First CSS Example</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

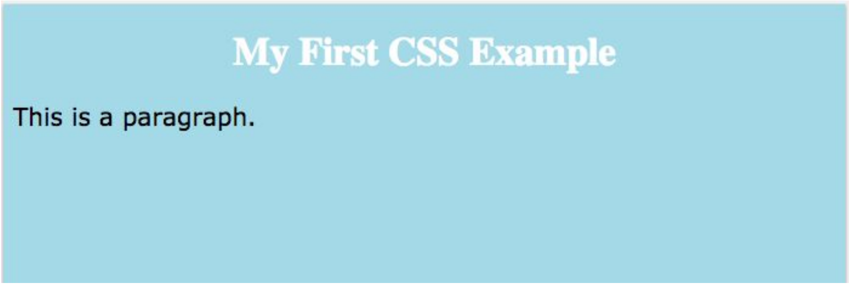


What Wins When Two Rules Conflict?

1. **Inline style** (`style="..."` on the element) — always wins over style sheets
2. **Internal and external style sheets** — no fixed ranking
    - When two rules target the **same element in the same way**, whichever comes **later in the `<head>`** wins
    - `<style>` then `<link>` → external wins
    - `<link>` then `<style>` → internal wins
2. **Browser default** — used only when nothing else applies

Example: ex1-css.html (internal style sheet)

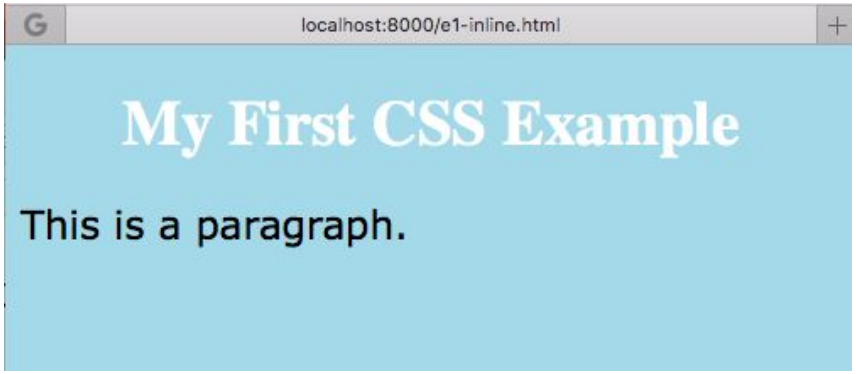
```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-color: lightblue;
}

h1 {
  color: white;
  text-align: center;
}

p {
  font-family: verdana;
  font-size: 20px;
}
</style>
</head>
<body>

<h1>My First CSS Example</h1>
<p>This is a paragraph.</p>

</body>
</html>
```



Example: CSS inline style

e1-inline.html (css) — Brackets

```
<!DOCTYPE html>
<html>
<body style="background-color:lightblue">
<h1 style="color:white;text-align:center;">My First CSS Example</h1>
<p style="font-family:verdana;font-size:20px;">This is a paragraph.</p>
</body>
</html>
```

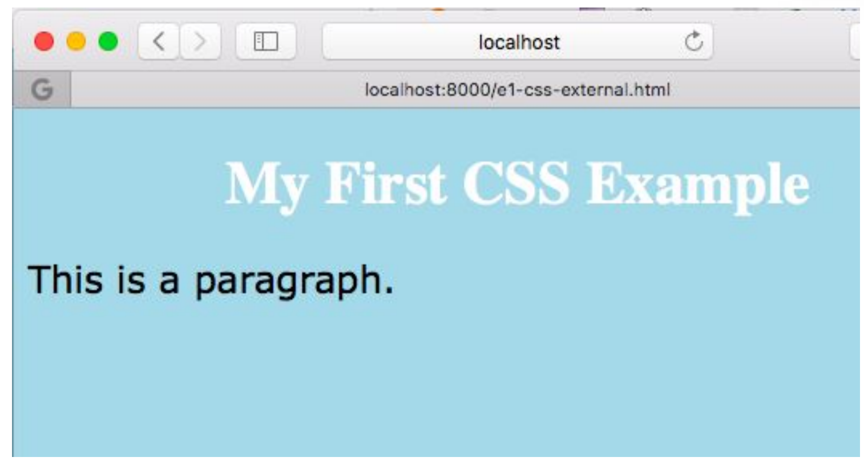


Example: External style sheet

```html
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet"
type="text/css" href="e1.css">
</head>
<body>

<h1>My First CSS Example</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

e1.css (css) — Brackets

```css
body {
  background-color: lightblue;
}

h1 {
  color: white;
  text-align: center;
}

p {
  font-family: verdana;
  font-size: 20px;
}
```



CSS Syntax

- The CSS syntax is made up of three parts
  - A selector: the HTML element/tag you wish to define
  - A property: the attribute you wish to change
  - A value: the value of the property
- Format
  - selector {property: value }

CSS Syntax Samples (1/2)

- The property and value are separated by a colon, and surrounded by curly braces:
  - body {color: black}
- If the value is multiple words, put quotes around the value
- If a font name contains spaces, put quotes around it

  p { font-family: "Times New Roman"; }
- Generic keywords (serif, sans-serif, monospace) are not quoted

  p { font-family: sans-serif; }
-

CSS Syntax Samples (2/2)

- If wish to specify more than one property, you must separate each property with a semicolon
  - p {text-align:center; color:red}
- To make the style definitions more readable, describe one property on each line
  - p {

    text-align:center;

    color:red

    }

Grouping

- You can group separators by separating each selector with a comma
- Example
  - All header elements will be displayed in green text color
  - h1, h2, h3, h4, h5, h6

```
{
    color:green
}
```

The class Selector (1/3)

- With the class selector you can define different styles for the same type of HTML element
- Example
  - Two types of paragraphs: one right-aligned paragraph and one center-aligned paragraph
  - p.right {text-align: right}
  - p.center {text-align: center}

The class Selector (2/3)

- You have to use the class attribute in your HTML document:
  - <p class="right"> This paragraph will be right-aligned. </p>
  - <p class="center"> This paragraph will be center-aligned. </p>
- To apply more than one class per given element, the syntax is:
  - <p class="center bold"> This is a paragraph. </p>

The class Selector (3/3)

- You can also omit the tag name in the selector to define a style that will be used by all HTML elements that have a certain class.
- In the example below, all HTML elements with class="center" will be center-aligned:
  - .center {text-align: center}
  - <h1 class=“center”>Center-aligned heading</h1>
  - <p class=“center”>Center-aligned paragraph</p>

Add Styles to Elements with Particular Attributes

- You can also apply styles to HTML elements with particular attributes.
- The style rule below will match all input elements that has a type attribute with a value of "text":
- Example
  - input[type="text"] {background-color: blue}

The id Selector (1/2)

- You can also define styles for HTML elements with the id selector. The id selector is defined as a #
- The style rule below will match the element that has an id attribute with a value of "green":
- #green {color: green}

The id Selector (2/2)

- The style rule below will match the p element that has an id with a value of ”left1":
- p#left1

 { text-align: center}

- Note: Do **NOT** start an ID name with a number! It will not work in Mozilla/Firefox.

The Descendant Selector

- To match an element **only when it is inside another element**, write the two selectors separated by a **space**.

```
selector1 selector2 { property: value }
```

→ matches every `selector2` that is inside a `selector1`

- .card h3 { color: crimson; }

Matches `<h3>` **only** inside an element with `class="card"`.

```
<div class="card">
  <h3>Styled in crimson</h3>     <!-- matched -->
</div>
<h3>Not styled</h3>              <!-- not matched -->
```

- It works at any depth: `.card p` matches a `<p>` inside a `<div>` inside a `.card` too.

Space vs Comma

- css

```css
/* <a> inside #navbar only */
#navbar a { color: white; }

/* the navbar itself AND every <a> on the page */
#navbar, a { color: white; }
```

- **Why use it?**
  Style an element by its context, without adding a class to every one.
- css

```css
.card p { font-style: italic; }   /* paragraphs inside cards */
p       { font-style: normal; }  /* all other paragraphs */
```

CSS Comments

- A comment will be ignored by browsers. A CSS comment begins with "/*", and ends with "*/“
- Example
  - /* This is a comment */

```
p { text-align: center;
/* This is another comment */
color: black;
font-family: arial }
```

External Style Sheet (1/2)

- An external style sheet is ideal when the style is applied to many pages
- With an external style sheet, you can change the look of an entire Web site by changing one file.
- Each page must link to the style sheet using the <link> tag. The <link> tag goes inside the head section

External Style Sheet (2/2)

- In an HTML page

```html
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" type="text/css" href="e1.css">
</head>
<body>
```

- In file e1.css

```css
body {
   background-color: lightblue;
}

h1 {
   color: white;
   text-align: center;
}

p {
   font-family: verdana;
   font-size: 20px;
}
```

Internal Style Sheet

- An internal style sheet should be used when a single document has a unique style.
- You define internal styles in the head section by using the <style> tag
- In an HTML page

```html
<html>
<head>
    <style>
        body {
            background-color: lightblue;
        }

        h1 {
            color: white;
            text-align: center;
        }

        p {
            font-family: verdana;
            font-size: 20px;
        }

    </style>
</head>
<body>
```

Inline Styles

- An inline style loses many of the advantages of style sheets by mixing content with presentation.
  - Use this method sparingly, such as when a style is to be applied to a single occurrence of an element
- Example

```
<body style="background-color:lightblue">
<h1 style="color:white;text-align:center;">My First CSS Example</h1>
<p style="font-family:verdana;font-size:20px;">This is a paragraph.
</p>
</body>
```

CSS Background

- The CSS background properties define the background effects of an element
- Some styles for background
  - Set the background color for an element
  - Set an image as the background

Example: Image as the Background

HTML Code

```
<html>
    <head>
        <style type="text/css">
        body
        {
        background-image:
        url('images/lela6.jpg')
        }
        </style>
    </head>
    <body>
        <h1>Two Languages to Learn</h1>
        <ul>
            <li>HTML</li>
            <li>JavaScript</li>
        </ul>
    </body>
</html>
```

HTML Presentation

CSS Text

- The CSS text properties define the appearance of text
- Some properties that we can set
  - Color
  - Alignment
  - Lowercase or uppercase

CSS Font Family

- CSS font properties define the font family, boldness, size, and the style of a text
- In CSS, there are two types of font family names:
  - Generic family - a group of font families with a similar look (like serif, or monospace)
  - Font family - a specific font family (like "Times New Roman" or "Arial")

Font Family Fallback List

- List several fonts separated by commas; the browser tries them left to right
- Always **end with a generic family** as a last resort

css

```css
h1 { font-family: "Times New Roman", Times, serif; }
p  { font-family: Arial, Helvetica, sans-serif; }
```

- Quote font names with spaces: `"Times New Roman"`
- Do **not** quote generic keywords: `serif`, not `"serif"`
- A quoted generic keyword becomes a search for a font with that exact name → the fallback silently fails

Example: Font Family

HTML Code

ex6.html (css) — Brackets

```html
<html>
<head>
<style type="text/css">
h3 {font-family: times}
p {font-family: courier}
p.sansserif {font-family: sans-serif}
</style>
</head>
<body>
<h3>This is header 3</h3>
<p>This is a paragraph</p>
<p class="sansserif">This is a paragraph</p>
</body>

```

HTML Presentation

localhost:8000/ex6.html

This is header 3

`This is a paragraph`

This is a paragraph

CSS Font Style

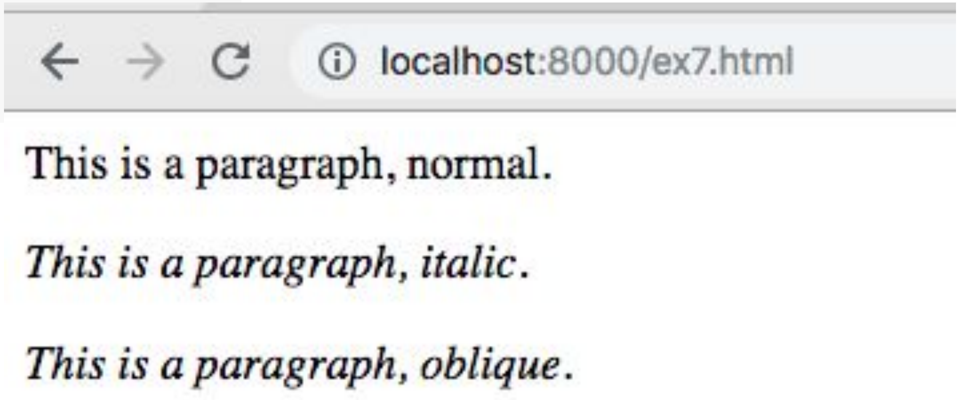
- The font-style property is mostly used to specify italic text
- This property has three values:
  - normal - The text is shown normally
  - italic - The text is shown in italics
  - oblique - The text is "leaning" (oblique is very similar to italic, but less supported)

Example: Font Style

- HTML Code

```html
<html>
<head>
<style type="text/css">
p.normal {font-style:normal}
p.italic {font-style:italic}
p.oblique {font-style:oblique}
</style>
</head>

<body>
<p class="normal">This is a paragraph, normal.</p>
<p class="italic">This is a paragraph, italic.</p>
<p class="oblique">This is a paragraph, oblique.</p>
</body>

</html>
```

- HTML Presentation



CSS Font Size

- Being able to manage the text size is important in web design
- However, you should not use font size adjustments to make paragraphs look like headings, or headings look like paragraphs.
- Always use the proper HTML tags, like <h1> - <h6> for headings and <p> for paragraphs.
- The font-size value can be an absolute, or relative size

CSS Font Absolute Size

- Sets the text to a specified size
- Does not allow a user to change the text size in all browsers (bad for accessibility reasons)
- Absolute size is useful when the physical size of the output is known

CSS Font Relative Size

- Sets the size relative to surrounding elements
- Allows a user to change the text size in browsers
- Remark If you do not specify a font size, the default size for normal text, like paragraphs, is 16px (16px=1em)
  - The em size unit is recommended by the W3C.

Example: Font Size

- HTML Code

```
<html>
<head>
<style type="text/css">
h1 {font-size: 150%}
h2 {font-size: 130%}
p {font-size: 100%}
.small-note { font-size: 0.8em; }
</style>
</head>
<body>
<h1>This is header 1</h1>
<h2>This is header 2</h2>
<p>This is a paragraph</p>
</body></html>
```

- HTML Presentation

This is header 1

This is header 2

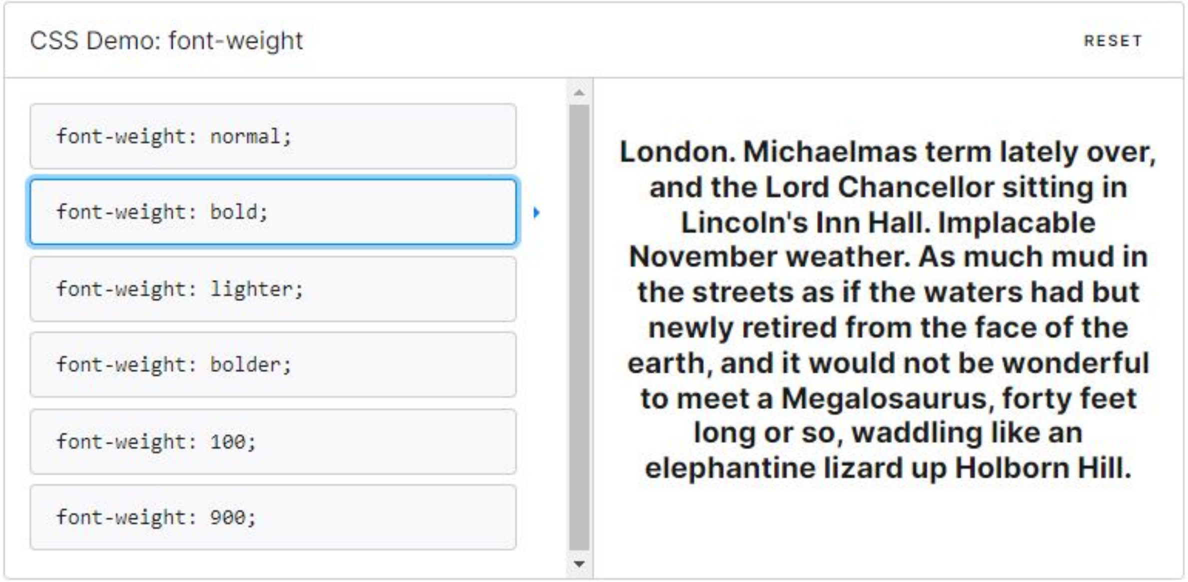
This is a paragraph

font-size property values

Value	Description
medium	Sets the font-size to a medium size. This is default
xx-small	Sets the font-size to an xx-small size
x-small	Sets the font-size to an extra small size
small	Sets the font-size to a small size
large	Sets the font-size to a large size
x-large	Sets the font-size to an extra large size
xx-large	Sets the font-size to an xx-large size
smaller	Sets the font-size to a smaller size than the parent element
larger	Sets the font-size to a larger size than the parent element
length	Sets the font-size to a fixed size in px, cm, etc. Read about length units
%	Sets the font-size to a percent of the parent element's font size

font-weight

- The font-weight CSS property sets the weight (or boldness) of the font
- The weights available depend on the font-family that is currently set.
- Try different font weights at font-weight - CSS: Cascading Style Sheets | MDN (mozilla.org)



CSS Border

- The CSS border properties allow you to specify the style and color of an element's border
  - The border-style property specifies what kind of border to display
  - The border-width property is used to set the width of the border
    - The width is set in pixels, or by using one of the three pre-defined values: thin, medium, or thick.

Example: Border Style and Width

```
<html><head><style type="text/css">
 p.one  {border-style:solid;border-width:5px;}
p.two { border-style:solid; border-width:medium;}
</style></head><body>
<p class="one">Some text.</p>
<p class="two">Some text.</p>
<p><b>Note:</b> The "border-width" property does not
work if it is used alone. Use the "border-style" property to
set the borders first.</p>
</body></html>
```

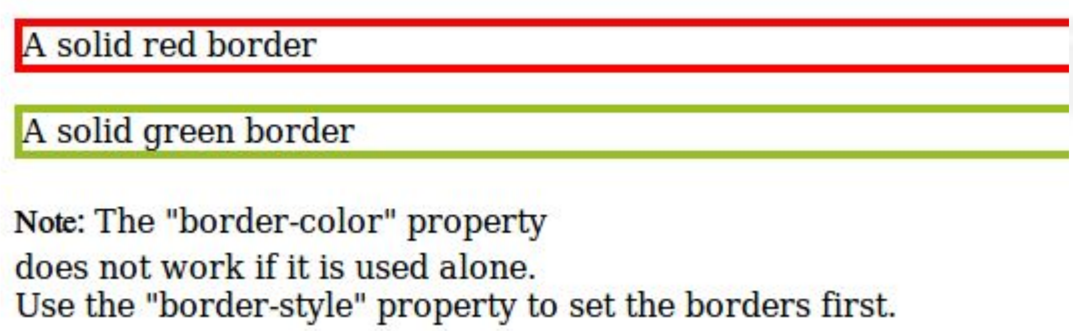
Some text.

Some text.

Note: The "border-width" property does not work if it is used alone. Use the "border-style" property to set the borders first.

Example: Border Color

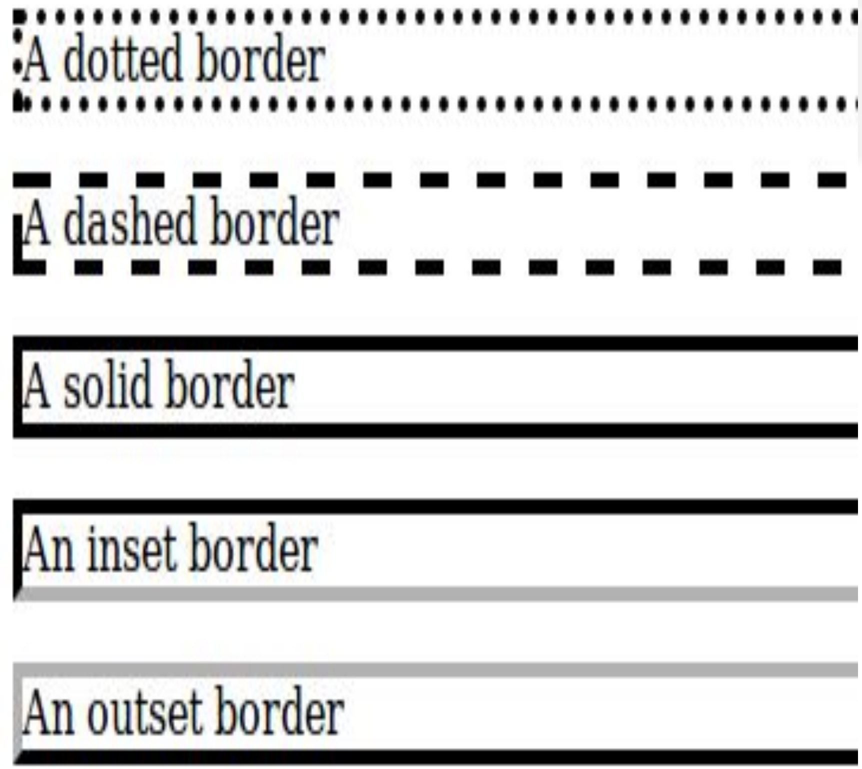
```
<html>
<head><style type="text/css">
p.one {
border-style:solid;
border-color:red;}
p.two {
border-style:solid;
border-color:#98bf21;
} </style>
</head>
<body>
<p class="one">A solid red border</p>
<p class="two">A solid green border</p>
<p><b>Note:</b> The "border-color"
property does not work if it is used alone.
Use the "border-style" property to set the
borders first.</p>
</body>
</html>
```



Example: Border Style

```
<html><head>
<style type="text/css">
p.dotted {border-style: dotted}
p.dashed {border-style: dashed}
p.solid {border-style: solid}
p.inset {border-style: inset}
p.outset {border-style: outset}
</style>
</head>
<body>
<p class="dotted">A dotted border</p>
<p class="dashed">A dashed border</p>
<p class="solid">A solid border</p>
<p class="inset">An inset border</p>
<p class="outset">An outset border</p>
</body>
</html>
```

- HTML Presentation



CSS Box Model

- All HTML elements can be considered as boxes. In CSS, the term "box model" is used when talking about design and layout
- In order to set the width and height of an element correctly in all browsers, you need to know how the box model works.
- The box model illustrates how the CSS properties: margin, border, and padding, affects the width and height of an element

CSS Box Model Layout

Box Model Explanation

- Content: The actual content of the box, where text and images appear.
- Padding: The space between the content and the border. It's part of the element's background.
- Border: The line that goes around the padding and content.
- Margin: The space outside the border, which separates the element from other elements

Width & Height of An Element

- When you specify the width and height properties of an element with CSS, you are just setting the width and height of the content area
- To know the full size of the element, you must also add the padding and border
- Margin is transparent space outside the element — add it to find the total space the element occupies on the page

Formulas for Computing Total Width and Total Height

- Total Width = Content Width + Left Padding + Right Padding + Left Border + Right Border
- Total Height = Content Height + Padding Top + Padding Bottom + Border Top + Border Bottom

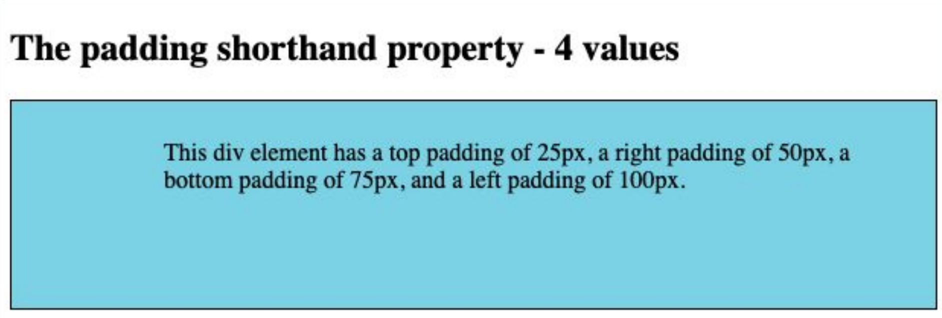
Width Box Model Example

- An Element
  - width:250px;
  - padding:10px;
  - border:5px solid gray;
  - margin:10px;
- What is a total width of this element?
  - 250 + 10*2 + 5*2 = 280

Setting Padding

- padding: <top> <right> <bottom> <left>
- padding: <top-bottom> <right-left>
- padding: <top> <right-left> <bottom>
- padding: <all-sides>

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  padding: 25px 50px 75px 100px;
  background-color: lightblue;
}
</style>
</head>
<body>

<h2>The padding shorthand property - 4 values</h2>

<div>This div element has a top padding of 25px, a right padding of
50px, a bottom padding of 75px, and a left padding of 100px.</div>
```



Setting Margin

Home › CSS › CSS Margins › Tryit: Margin shorthand property - 3 values

Run ❯ Result Size: 625 x 430 Get your own website

```html
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin: 25px 50px 75px;
  background-color: lightblue;
}
</style>
</head>
<body>

<h2>The margin shorthand property - 3 values</h2>

<div>This div element has a top margin of 25px, a right and left
margin of 50px, and a bottom margin of 75px.</div>
```

The margin shorthand property - 3 values

This div element has a top margin of 25px, a right and left margin of 50px, and a bottom margin of 75px.

Example: Box Model

HTML Code

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML
    1.0 Transitional//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xht
    nsitional.dtd">
<html><head>
<style type="text/css">
div.ex
{width:220px;
padding:10px;
border:5px solid gray;
margin:0px;}
</style></head><body>
<div class="ex">The line above is 250px wid
    />
Now the total width of this element is also
    250px.</div>
<p><b>Note:</b> In this example we have a
    DOCTYPE<br/>
 declaration (above the html element),<br/>
so it displays correctly in all browsers.</p>
</body>
</html>
```

HTML Presentation

The line above is 250px wide.

Now the total width of this element is also 250px.

Note: In this example we have added a DOCTYPE declaration (above the html element), so it displays correctly in all browsers.

CSS List

- In HTML, there are two types of lists:
  - unordered list - the list items are marked with bullets (typically circles or squares)
  - ordered list - the list items are marked with numbers or letters
- With CSS, lists can be styled further, and images can be used as list item markers.

Example: List

```html
<html><head><style type="text/css">
ul.circle {list-style-type:circle}
ul.square {list-style-type:square}
ol.upper-roman {list-style-type:upper-roman}
ol.lower-alpha {list-style-type:lower-alpha}
</style></head><body>
<p>Type circle:</p>
<ul class="circle"><li>Coffee</li>
<li>Tea</li></ul>
<p>Type square:</p>
<ul class="square"><li>Coffee</li>
<li>Tea</li></ul>
<p>Type upper-roman:</p>
<ol class="upper-roman"><li>Coffee</li>
<li>Tea</li></ol>
<p>Type lower-alpha:</p>
<ol class="lower-alpha"><li>Coffee</li>
<li>Tea</li></ol>
</body></html>
```

Type circle:

- Coffee
- Tea

Type square:

- Coffee
- Tea

Type upper-roman:

I. Coffee
II. Tea

Type lower-alpha:

a. Coffee
b. Tea

References

- W3Schools, “CSS Tutorial”, available at http://www.w3schools.com/css/default.asp
- Anthropic. Claude AI [Internet]. San Francisco: Anthropic; 2024 [cited 2024 Jun 30]. Available from: https://www.anthropic.com